

CROP SCIENCE SEMINAR BY SIR WELLUS in line with BOLZANO WIERSTRASS RULES

RESPIRATION

Cellular respiration is the process by which the energy contained in organic molecules is made available for all of the active processes within a cell.

The process of respiration can occur with oxygen (aerobic) or without oxygen (anaerobic)

Aerobic respiration can be divided into four stages:

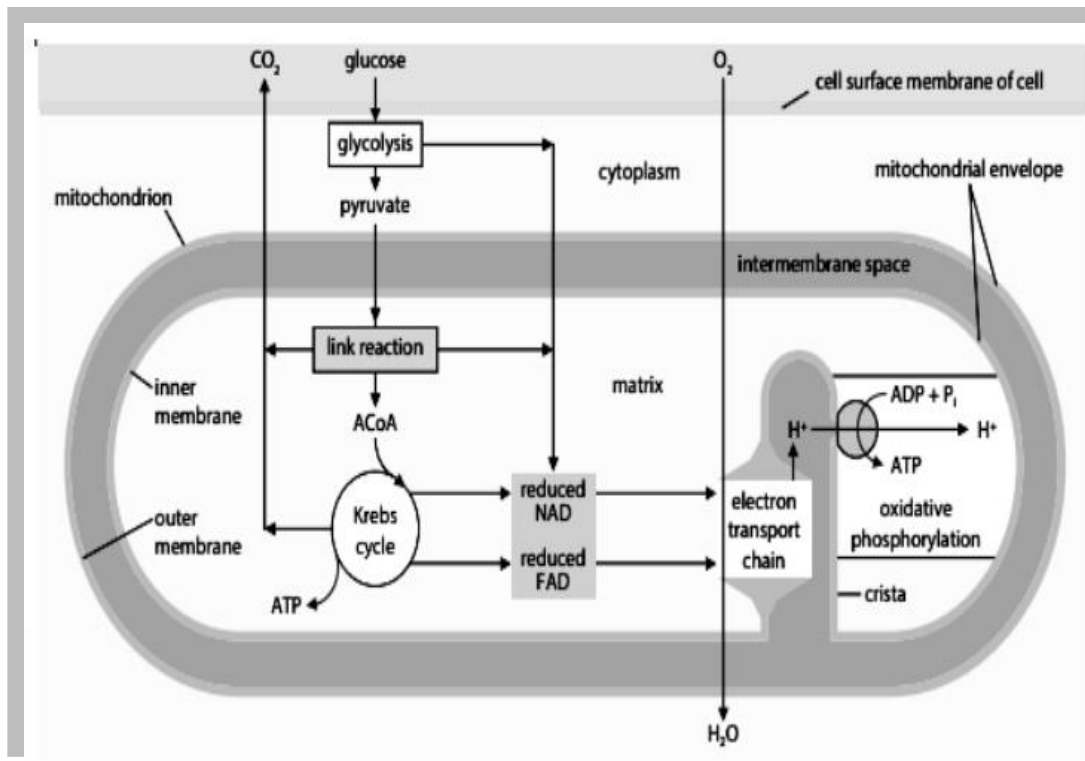
1. Glycolysis (G) ---Glucose is converted to pyruvic acid. Hydrogen is removed and is passed to the electron carriers

2. The Link reaction (LR) ----Pyruvate enters mitochondrion, is decarboxylated, dehydrogenated and combines with coenzyme A to give acetyl coenzyme A. The hydrogen which is removed is passed to the electron carriers.

3. Kreb's cycle (K)---- A cyclical series of reactions during which hydrogen is passed to the electron carriers, carbon dioxide is removed and the starting reagents are regenerated.

4. The electron transfer chain (ETC).....-- The hydrogen from glycolysis and Kreb's cycle is split to release electrons. These pass through carriers and generate ATP. The hydrogen reforms and is combined with oxygen to release water.

DIAGRAM



Glycolysis

- occurs in the cell **cytoplasm** in both aerobic and anaerobic conditions..

- Glycolysis is the first stage of respiration

The steps in glycolysis

1. Phosphorylation: Glucose is phosphorylated, i.e. ATP is used to add a phosphate group to glucose. This makes the glucose more reactive, allowing it to be broken down. A six carbon (6C) phosphorylated sugar is produced called hexose 1,6-bisphosphate.

Glucose + 2ATP → Fructose 1,6-bisphosphate

2. Lysis: The hexose 1,6-bisphosphate (6C) is split into two molecules of a 3C sugar phosphate called Triose Phosphate (TP) (aka glyceraldehyde-3-phosphate abbreviated as TP, G3P or GALP).

Fructose 1,6-bisphosphate → 2 Triose phosphate

3. Oxidation: Hydrogen is removed from each of the TP molecules, i.e. the TP molecules are oxidised. The hydrogen is passed to NAD, a coenzyme, which is reduced.

4H + 2NAD → 2NADH + 2H⁺

Dephosphorylation: phosphates are transferred from the intermediate substrate molecules to form 4 molecules of ATP through substrate-linked phosphorylation



Production of pyruvate: The removal of hydrogen from TP produces pyruvic acid (PA) (aka pyruvate).



LINK REACTION

PA enters the mitochondrion

Carbon dioxide and hydrogen are removed from the PA by decarboxylase and dehydrogenase enzymes respectively to produce an acetyl group, $\text{CH}_3\text{C}(\text{O})^-$.

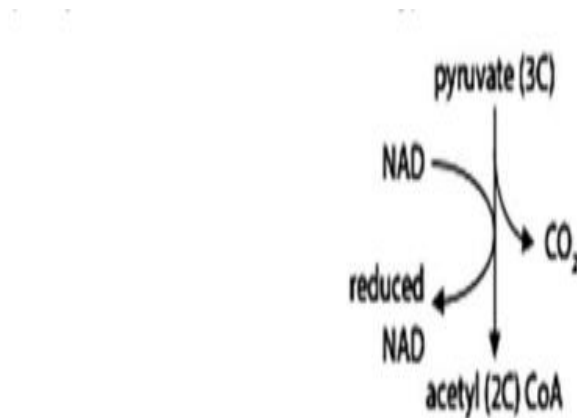
The acetyl group is combined with coenzyme **A (CoA)** to form a **2C compound called acetylcoenzyme A (AcCoA)**.

Pyruvate + NAD + CoA $\rightarrow$ acetyl CoA + carbon dioxide + reduced NAD

The link reaction produces:

- **Acetyl coA**
- **Carbon dioxide (CO_2)**
- **Reduced NAD (NADH)**

The Link Reaction - occurs only if oxygen is available



Kreb's Cycle - only occurs if oxygen is available.

The carbon dioxide is released. The hydrogen which is removed is passed to coenzymes such as NAD and FAD (i.e. the coenzymes are reduced).

Eventually the 4C compound (oxaloacetate) is regenerated. This then combines with more AcCoA and the whole cycle begins again.



Each hydrogen atom is split into a hydrogen ion (H^+) and an electron (e^-).

As the electron passes **down from one electron carrier to another, some of their energy is released**. This energy is **used to convert ADP into ATP, i.e. to phosphorylate ADP**. Since the removal of hydrogen or electrons from substance is defined as oxidation, the overall reactions of the ETC are known **as oxidative phosphorylation**.

Finally, each electron is reunited with a hydrogen ion (H^+) and the hydrogen which forms immediately combines with oxygen to form water

The Role of NAD & FAD

- Coenzymes NAD and FAD play a critical role in aerobic respiration
- When hydrogen atoms become available at different points during respiration NAD and FAD accept these hydrogen atoms

Importance of ATP

ATP consists of the organic base adenine, the 5-carbon sugar ribose and three phosphate groups. This ATP is used for a variety of processes.

- **Anabolic processes.** ATP provides the energy necessary to build up macromolecules, e.g. proteins from amino acids.
- **Active transport.** Energy is needed to move materials against a concentration gradient, e.g. by the use of ion pumps.
- **Movement.** Contraction the spindle fibres during cell division requires energy from the hydrolysis of ATP.
- **Activating reactants.** Chemicals often require the addition of phosphate groups from ATP to make them more reactive, e.g. phosphorylation of glucose at the beginning of glycolysis.
- **Secretion.** ATP provides energy for the secretion of cell products.

TUTORIAL QUESTIONS

Outline the role of ATP plants.

Where does the following processes occur

(i) Glycolysis

(ii) Kreb's cycle

Briefly describe the processes of glycolysis

Define the following terms

(i) phosphorylation

ii) carboxylation

iii) hydrogenation

a) Explain why ATP is needed at the start of glycolysis. (1)

b) State the role of NAD in glycolysis. [1]

c) State how many molecules of ATP are produced from one molecule of glucose during glycolysis. [1]

d) Name the two types of reaction that occur during the conversion of pyruvate to acetyl CoA in the link reaction. [2]

e) Name the location, in the mitochondrion, of the link reaction. [1]

f) Describe what happens to the hydrogen released during the link reaction